# MOHAMMAD OTAQI

## SOFTWARE ENGINEER

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## SUMMARY


Software Engineer at ZagTrader building capital-markets infrastructure: SWIFT (ISO 15022) securities settlement and its public REST API, FIX connectivity, exchange and market-data connectors, cryptographic licensing, and an AI-driven test-automation platform. Works across C#/.NET 10, ASP.NET Core, TypeScript (React, Node/Fastify), SQL Server and PostgreSQL, with Docker-based CI/CD to production. Also builds and ships his own mobile apps and multi-tenant SaaS products end to end with Flutter, React Native, Next.js and Django. Google-certified in IT Support and Project Management.


## EDUCATION

**Al Hussein Technical University (HTU)** **Oct 2021 – July 2025**

- Bachelor in Computer Science
- GPA: Excellent

**Jubilee Institute**

- Organized Sports Day event as part of school leadership activities.
- Participated in Line Robotics Competition (5th place).
- Engaged in extracurricular activities including Model United Nations (MUN) and volunteering programs.

## PROFESSIONAL EXPERIENCE

**Software Engineer — ZagTrader** **September 2025 – Present**

- Built the Swift Suite (Engine, Bridge, Monitor): end-to-end SWIFT ISO 15022 securities settlement (MT540–548, MT564, MT598/599) with maker-checker approval — C#/.NET 10.
- Engineered the Swift Engine pipeline (reader, validator, builder, archiver) with MT validation via Prowide/IKVM.
- Developed the Swift Bridge gateway to Fineksus, Tadawul/Edaa and SFTP; rebuilt Tadawul/Edaa templates from the ISO 15022 spec and fixed live network NAKs.
- Designed a public REST API for the platform (55 OpenAPI operations): scoped, rotatable API keys, idempotent submission, signed webhooks with an event outbox; 59/59 endpoints verified live.
- Added Azure Key Vault / HashiCorp Vault secrets management, an RBAC permission system and offline TOTP 2FA across the suite.
- Built the GUI and production release pipeline for ZT-AMTP, a multi-agent LLM platform that crawls web apps and generates, runs and certifies Playwright tests (Temporal, Fastify, Drizzle/PostgreSQL, React); raised the customer-app pass rate from 41% to 67.7%.
- Delivered FixPortal to production: QuickFIX/n dictionary validation, per-market rules and message diffing — ASP.NET Core, EF Core, PostgreSQL, React; 170+ automated tests and CI/CD.
- Integrated exchange connectivity: Crypto.com FIX (HMAC logon, TLS pinning, market data + order entry), AIX FIX reference data, BVB SOAP reference data (845 instruments), CoinMENA REST with Ed25519 signing.
- Added CMC Markets (FX), OKX and IDC feed readers; built the IDC historical reader (.NET 10, WebSocket → SQL Server) that replaced a legacy PHP job.
- Created WatchList dashboard (React) with real-time SignalR market-data visualization.
- Shipped ZagKeyGenWeb crypto licensing (AES/HMAC/TripleDES) to production on Docker Compose with PostgreSQL and an audit trail.
- Migrated five Swift/FIX repositories from SVN to Gitea; analyzed ZagTraderConnector integrations and database architecture; self-contained apps via Inno Setup & batch.

**Flutter & Mobile Application Developer — Lightning Gate (JPECO)** **July 2024 – March 2025**

- Developed and maintained mobile applications using Flutter & Firebase, ensuring stability and scalability.
- Partnered with developers and designers to implement features and fix bugs, improving user experience and app reliability.

**Technical Support System Controller — HTU Student Activities** **November 2021 – July 2024**

- Managed IT systems and coordinated technical support for student events and activities.
- Led technical operations for large-scale events, including MACH1 with AreoSquad at King's Academy.

**Esports Staff — Jordan Esports Federation (JEF/JEG)** **July 2023 – May 2025**

- Provided IT support and served as a referee during official esports tournaments.
- Ensured smooth execution of competitive events by managing technical infrastructure.

**Technical Support and Web Development — HTU Online Learning Center** **June 2023 – Oct 2023**

- Contributed to the HTUx platform, drafting user manuals and technical documentation to streamline onboarding.
- Delivered IT support to staff, reducing issue resolution time and improving workflow continuity.

## SKILLS

**Technical Skills:**

- **Programming:** C, C#, .NET, ASP.NET Core, React, TypeScript, Node.js, Flutter, React Native, Firebase, Python, Java, JavaScript, Dart, HTML/CSS, SQL.
- **Backend & APIs:** ASP.NET Core, EF Core, Fastify, Zod, Drizzle ORM, NestJS, Prisma, Django/DRF, REST/OpenAPI, Webhooks, SignalR.
- **Software Engineering:** Algorithms, Data Structures, OOP, Multi-tenant SaaS, Operating Systems.
- **Data & Analytics:** SQL Server, PostgreSQL, Supabase, Valkey/Redis, Power BI, Tableau, NumPy, pandas, Excel.
- **Tools & Platforms:** Git/GitHub, Gitea, SVN, Docker Compose, CI/CD (GitHub/Gitea Actions, EAS), Linux servers, IIS, Visual Studio, VS Code, Inno Setup, Android Studio, Figma, Justinmind, Adalo.
- **Security & Protocols:** SWIFT (ISO 15022/MT), FIX 4.4/5.0 SP2 (QuickFIX/n), SoupBinTCP, ITCH, SOAP, Cryptography (AES/HMAC/Ed25519), TLS pinning, JWT, OAuth/Entra SSO, TOTP 2FA, RBAC, Azure Key Vault, HashiCorp Vault, IKVM.
- **Testing & Automation:** Playwright, Vitest, Jest, xUnit, Temporal, LLM test agents.
- **IT Support:** Troubleshooting, Networking basics, Documentation, System Maintenance.
- **AI & Local LLMs:** Ollama, LM Studio, llama.cpp/GGUF (K-quants, layer offload), CUDA, GPU/VRAM sizing, PC builds, Hugging Face (Llama/Qwen/Mistral).
- **AI Engineering:** prompt & context engineering, structured/JSON output, tool/function calling, agent harnesses, MCP (Model Context Protocol), Claude Code/Cursor.

**Professional & Soft Skills**

- Problem Solving & Critical Thinking
- Time Management & Organization
- Leadership & Teamwork
- Communication (Tech & Non-Tech)
- Project Coordination & Multitasking
- Adaptability & Flexibility
- Curiosity & Continuous Learning
- Analytical Thinking & Decision-Making
- Conflict Resolution & Negotiation
- Client & User Support

**Languages:**

- Arabic — Native
- English — Fluent (C1)
- French — Beginner (Delf A1)

## LEADERSHIP & VOLUNTEERING

- ACM Treasurer & IT Support — Managed finances and maintained IT systems.
- GDVC IT Support — Provided event support; awarded Best Member (2024).
- Students Deanship Volunteer — Supported campus-wide student initiatives.
- Jubilee Institute — Organized sports day; earned 5th place in Line Robotics.
- Volunteered with MUN, Run Jordan, and other community events.

## CERTIFICATIONS & AWARDS

- Google (Coursera): IT Support (2023), Project Management (2023)
- DataCamp: Python, NumPy, pandas, Data Manipulation with pandas
- Award: Best Member — GDVC (2024)
- Top 24 Finalist — Ahli Fintech Hackathon (2025)
- 7th Place — Tech 3arabi Hackathon (2025)

## INTERESTS

- Artificial Intelligence & Machine Learning
- Augmented & Virtual Reality (AR/VR)
- Game Development
- Computer Architecture
- Hackathons & Tech Competitions

## PROJECTS

**Sudur — Quran Memorization App with AI Recitation Grading** (began as Sabaq, team hackathon project) **| React Native, Expo, Django, DRF, PostgreSQL**

- Started as Sabaq with a 3-developer team; owned the Django/DRF backend, the OAuth2 proxy and MCP integration to the Quran Foundation API.
- Built AI recitation grading — Whisper speech-to-text fuzzy-matched against riwaya-correct text with diff feedback.
- Led the release as Sudur: Google Play closed testing and App Store submission, Arabic + English, website at sudur.app.
- Gesture-based Mushaf reader, verse-by-verse audio, prayer times with adhan, revision scheduling and Android home-screen widgets.
- Release pipeline: push to main → 229 Jest tests → signed build → Play testing track with release notes; RevenueCat subscription, AdMob with UMP consent.

**Wared (وارد) — Inventory & Warehouse Management SaaS | Django 5, DRF, PostgreSQL, django-tenants, React, TypeScript**

- Built a schema-per-tenant SaaS for Jordanian distributors and wholesalers with an Arabic-first, RTL React PWA and barcode scanning.
- Append-only stock ledger with exact decimal money (JOD, 3 dp) and row-locked balances; weighted-average costing, FEFO expiry, serials and landed cost.
- Purchasing, sales, put-away, stocktakes, supplier returns and proof of delivery; JoFotara e-invoicing with credit notes.
- Plan tiers, owner console, per-tenant subdomains with on-demand TLS, and backups proven by a real restore.
- Implemented and tested all 45 Phase-1 "must" requirements; dedicated review agents gate tenant isolation, permissions, money arithmetic and migrations.

**MOT — Gamer Social Network & LFG Matchmaking** (Google Play internal testing) **| Flutter, Firebase, Riverpod, RevenueCat**

- Built a cross-platform app for gamers to find teammates and connect — profiles plus look-for-group matchmaking by game, platform and playstyle.
- Implemented real-time 1:1 chat with typing/seen indicators and FCM push notifications.
- Integrated a live gaming-news feed and a 300+ game database via the RAWG API.
- Added content moderation (toxicity filtering, reporting, age-gating) and Cloud Functions for feed fan-out and notifications.
- Shipped premium tiers and an in-app store with RevenueCat; analytics and crash reporting via Firebase.

**Mabsim — Multi-Tenant Clinic Management SaaS | Next.js 16, TypeScript, Supabase, PostgreSQL**

- Built a clinic platform for MENA practices: appointments, online booking with phone verification, patient records, invoices, recalls and waitlist.
- Specialty records: dental odontogram and a 44-point orthopedic chart; insurance claims, lab cases, prescriptions and consent letters.
- Integrated JoFotara (Jordan's national e-invoicing) and WhatsApp reminders via the Meta Cloud API behind a per-clinic sending gate.
- Tenant isolation with Postgres row-level security; an exclusion constraint blocks overlapping bookings, a unique index double invoices, a trigger overpayment.
- Owner console with subscription warnings and one-click suspension; encrypted backups with a tested restore drill; Arabic-first RTL.

**Word Oasis (واحة الكلمات) — Arabic Word Puzzle Game | Flutter, Dart, AdMob, RevenueCat, Python**

- Built an Arabic letter-wheel crossword with 827 levels across 18 landmark chapters.
- Wrote a Python level builder that generates the levels from a 50,000-word Arabic list.
- Daily challenge, lucky wheel, daily gift, chapter treasures, hints and a coin economy.
- Monetization: AdMob banner, interstitial, rewarded and app-open ads with UMP consent; RevenueCat in-app purchases.
- Prepared the store release (signing, listing, graphics, content rating) and submitted it to Google Play closed testing.

**Odispear — Discord-style Community Platform** (Live · n0tmot.com) **| React, TypeScript, Node.js, PostgreSQL, Redis, LiveKit, AWS**

- Built a full-stack community platform — servers, channels, roles and direct messages — with real-time messaging over Socket.IO.
- Integrated LiveKit voice/video with RNNoise/Krisp noise filtering, push-to-talk and voice-activity detection.
- Implemented JWT + Google OAuth, Stripe subscription billing, and AWS S3 media on a PostgreSQL + Redis backend.
- Deployed first to AWS EC2 behind nginx with PM2, now on Docker Compose behind Caddy. Live at n0tmot.com.

**MedGuard — Health Vitals Companion** (Graduation Project, Prototype) **| Flutter, Firebase (Auth/Firestore), BLE**

- Built a mobile app that lets a Care Receiver share vitals (heart rate, $SpO_2$, stress) with a Carer in real time via Firestore.
- Implemented email/password auth with password reset and role-based views (Care Receiver / Carer).
- Added optional BLE heart-rate read (GATT 0x180D/0x2A37) and a mock device scan flow.
- Built a GPS emergency-alert pipeline that runs AI health analysis (Gemini Cloud Function) and pushes critical alerts to the Carer over FCM, with condition-based first-aid tips.
- Structured code with screens/widgets, state via provider; documented setup, permissions, and Firebase rules.

**Masra — Islamic Education Platform** (Flutter app + Node API) **| Flutter, Node.js, Express, Prisma, PostgreSQL, Supabase**

- Built a Flutter app (offline-first caching, Arabic/English localization) for news, fatwa search, courses and gamified 30-day challenges.
- Developed a Node.js/Express + Prisma API on PostgreSQL (Supabase) with row-level security and a secure points RPC.
- Created a 12-panel admin CMS for managing all content — news, fatwas, courses, challenges and users.
- Set up GitHub Actions CI/CD for linting, builds and automated APK releases.

**Sehleh — On-Demand Local-Services Marketplace | Flutter, BLoC, Supabase, PostGIS, Next.js, Stripe, Tap**

- Built customer and provider Flutter apps (BLoC) for booking local services, with live provider GPS tracking and in-app chat.
- Implemented geolocation discovery using PostGIS spatial queries and multi-role access (customer / provider / admin).
- Integrated Stripe and Tap payments via Supabase Edge Functions, plus FCM push notifications and KYC verification.
- Built a Next.js admin dashboard for users, providers, bookings and promotions.

**onecli — Multi-Model AI Coding CLI | Node.js**

- Built a zero-dependency CLI that wraps the Claude, Codex, Gemini and Qwen CLIs behind one streaming REPL with mid-chat model switching.
- Shared project "brain" that syncs context files across tools; planner → workers → auditor orchestration pipeline.
- "Council" mode where several models answer and one judges; local web dashboard and Windows desktop control.

**Tabayyun — AI Arabic News & Image-Tampering Verifier | React, Vite, OpenAI, Google GenAI, Sightengine**

- Built an Arabic-first web tool that scores news credibility from a URL or text using LLM analysis plus keyword/sentiment heuristics.
- Implemented image forensics — EXIF metadata, Error-Level Analysis and Sightengine AI-generated-image detection.
- Added a trusted-source database and reverse-image-search links; designed a fully right-to-left, animated UI.
- Hackathon project (Tech 3arabi).

**FactQuest — Gamified Media-Literacy Trainer | React, Vite, Tailwind CSS**

- Built an interactive web app that teaches users to spot misinformation, deepfakes and bias.
- Designed eight challenge modules — deepfake, clickbait, bias, sponsored, satire, image and source verification — with guided lessons.
- Implemented XP, streaks and level progression with an animated mascot for feedback.
- Hackathon project.